

# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Developed production-grade agentic systems using **Python**, implementing stateful orchestration with **LangGraph** to manage complex, multi-step workflows for mission-critical operations, achieving a **99.9% uptime** SLA.
- Engineered context management strategies across long-running agent executions, integrating **MCP-style tools** and ensuring state consistency for over **10,000 daily transactions** processed by AI agents.
- Built and maintained **RAG pipelines**, leveraging structured, semi-structured, and unstructured data sources, incorporating advanced retrieval strategies and embeddings to enhance AI model accuracy by **18%**.
- Designed and implemented context management strategies for multi-step and long-running agent executions, reducing processing errors by **22%** through improved state tracking.
- Optimized **Python** application performance for AI agent systems, reducing average execution time by **30%** and improving overall system responsiveness in a highly regulated environment.
- Supported the integration of AI agentic systems with existing enterprise infrastructure, ensuring seamless data flow and operational continuity across **relational databases** and **NoSQL vector stores**, contributing to a **15%** reduction in manual processing time.

## SKILLS

**Core Technologies** – Python, LangGraph, RAG, AI, Agentic Systems, SQL, NoSQL.

**Data Management** – Vector Stores, Embeddings, Data Pipelines, ETL, Data Warehousing.

**Cloud & DevOps** – AWS, Docker, Jenkins, Git, Linux.

**Programming Languages** – Python, Java, Spring Boot, R.

**Databases** – PostgreSQL, MySQL, Oracle, Vector Databases.

**Frameworks & Libraries** – LangGraph, Pandas, NumPy, Scikit-learn.

**Observability & Monitoring** – Splunk, Dynatrace, CloudWatch.

## EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of  
Ireland, Dublin**

**02/2026**

**BACHELOR OF ENGINEERING in Computer Science – SNS College  
of Technology, Coimbatore, India**

**01/2021**

## PROJECTS

**Production-Grade AI Agentic Workflow Orchestrator (HCL Technologies)**

- Architected and deployed a **Python**-based agentic system for a financial services client, managing complex transaction processing workflows using **LangGraph** for stateful orchestration.
- Implemented sophisticated context management and RAG architectures, integrating diverse data sources and optimizing retrieval strategies to achieve a **99.95% accuracy** rate in anomaly detection.

- Developed automated monitoring and alerting mechanisms with **Splunk** and **CloudWatch**, ensuring **99.9% availability** and supporting regulatory compliance through detailed audit trails.

#### **Power BI Executive Dashboard Suite (MSc Data Analytics Project)**

- Designed and delivered an interactive suite of **Power BI** dashboards, consolidating KPIs from multiple enterprise systems for executive review, improving data accessibility for stakeholders.
- Developed complex **DAX** measures and optimized data models within **Power BI**, reducing report load times by **40%** and enabling real-time data exploration.
- Automated data extraction and transformation processes using **Power Query** and **SQL**, ensuring data integrity and timely updates for all published reports.

### **ACHIEVEMENTS & CERTIFICATIONS**

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)